

e-CAM_YUV-GMSL

Getting Started and Developer Guide



e-con Systems

Think Camera. Think e-con.

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Disclaimer

The specifications of e-CAM_YUV-GMSL-PRODUCTS board and instructions on how to use this board with Jetson AGX Orin™ development kit is provided as reference only and e-con Systems reserves the right to edit/modify this document without any prior intimation of whatsoever.

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Introduction to NileCAM

The NileCAM camera module from e-con Systems with high-speed unified serial interface that carries video data, control data and power in a single coaxial cable. This board has been designed for the NVIDIA® Jetson Orin™ development kit.

e-CAM_YUV-GMSL-PRODUCTS uses serializer-deserializer (SerDes) technology that allows single coaxial cable interface for the camera. With coaxial cable lengths of up to 15m, The fleet cameras are fitted on trucks for scene recognition, driver guidance and smart city applications including outdoor surveillance, smart parking, smart vehicles, and kiosks.

This e-CAM_YUV-GMSL_PRODUCTS supports the following list of products STURDeCAM20_CUOAGX, STURDeCAM21_CUOAGX, NileCAM20_CUOAGX, NileCAM21_CUOAGX, NileCAM25_CUOAGX and NileCAM81_CUOAGX.

The flexibility of coaxial cable and the ability to carry video data, control data and power in a single cable makes e-CAM_YUV-GMSL-PRODUCTS fast (low latency), flexible and yet reliable. e-CAM_YUV-GMSL-PRODUCTS uses Maxim's Gigabit Multimedia Serial Link (GMSL) interface technology for the serial interface to the camera. This serial interface uses coaxial cable that carries high-speed video data from the camera, bidirectional control data between the camera and host controller, and power supply for the camera. The power to the camera is supplied from the host processor through this coaxial cable.

e-CAM_YUV-GMSL-PRODUCTS is a GMSL camera contains a CMOS image sensor from onsemi™ along with on-board Image Signal Processor (ISP). The coaxial cable is connected to camera module enclosure through a rugged FAKRA connector. e-CAM_YUV-GMSL-PRODUCTS is interfaced to the J509 connector on the Jetson AGX Orin™ using the e-CAM_YUV-GMSL-PRODUCTS deserializer board.

The NVIDIA® Jetson AGX Orin™ developer kit is a full-featured development platform for visual computing. It is ideal for applications requiring high computational performance in a low power envelope. The Jetson AGX Orin™ developer kit is pre-flashed with a Linux environment which includes support for many common APIs. This kit is supported by NVIDIA® complete development tool chain.

e-con Systems provides the sample applications that demonstrate the features of this camera. However, this camera can also be utilized by any V4L2 application.

This document describes how to interface e-CAM_YUV-GMSL-PRODUCTS board on Jetson AGX Orin™ development kit and to check the image preview.

Prerequisites

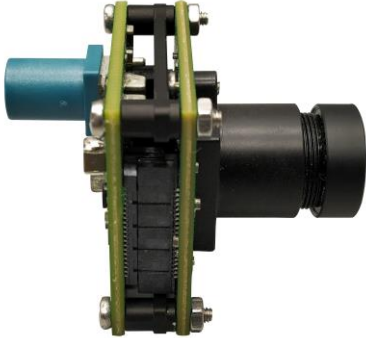
The prerequisites are as follows:

- A Host PC running Ubuntu 22.04/20.04 Operating System
- Necessary accessories such as keyboard, mouse, HDMI monitor and so on, are required to be connected to NVIDIA® Tegra® development kit.
- A Jetson AGX Orin™ development kit pre-flashed with L4T36.3.0

Parts Supplied

The following table lists the parts supplied with the kit.

Table 1: Parts Supplied and its Quantity

Parts Supplied	Image	Quantity
e-CAM_YUV-GMSL-PRODUCTS camera module with lens fixed (Reference image)		X
Deserializer board		1
Coaxial cable (3m/15m length)		X
12V 2A power adapter for the deserializer board		1

Screws (deserializer to Orin kit)		3
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Note: Quantity X depends on the part number ordered as listed in below table.

Table 2: Parts Numbers and its Quantity

Part Number	X
e-CAM_YUV-GMSL-PRODUCTS_1H02R1	1
e-CAM_YUV-GMSL-PRODUCTS_6H02R1	6

Description

The e-CAM_YUV-GMSL-PRODUCTS board is designed to interface one or six 2 MP or 2.3 MP or 8 MP camera modules with the Jetson AGX Orin™ development kit. e-CAM_YUV-GMSL-PRODUCTS has two parts. The first part is the e-CAM_YUV-GMSL-PRODUCTS camera module containing the image sensor-based camera module from onsemi™, along with ISP. The second part is the deserializer board for interfacing with Jetson AGX Orin™ development kit.

e-CAM_YUV-GMSL-PRODUCTS has below listed parts:

- e-CAM_YUV-GMSL-PRODUCTS camera module
- e-CAM_CUOAGX_DESER_6 (deserializer Board)

The e-CAM_YUV-GMSL-PRODUCTS camera module and deserializer board are shown in below figures.



Figure 1: e-CAM_YUV-GMSL-PRODUCTS Camera Module Board

(Reference image)

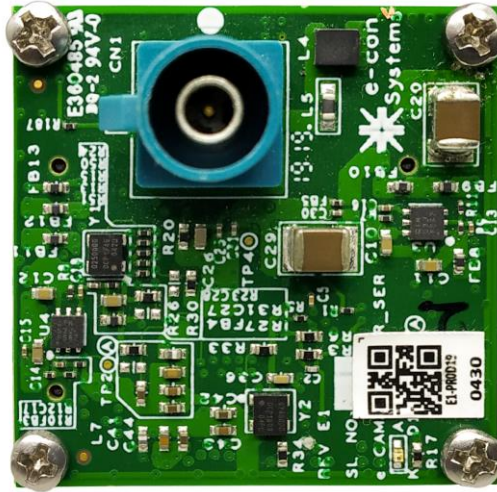


Figure 2: e-CAM_YUV-GMSL-PRODUCTS Serializer Board

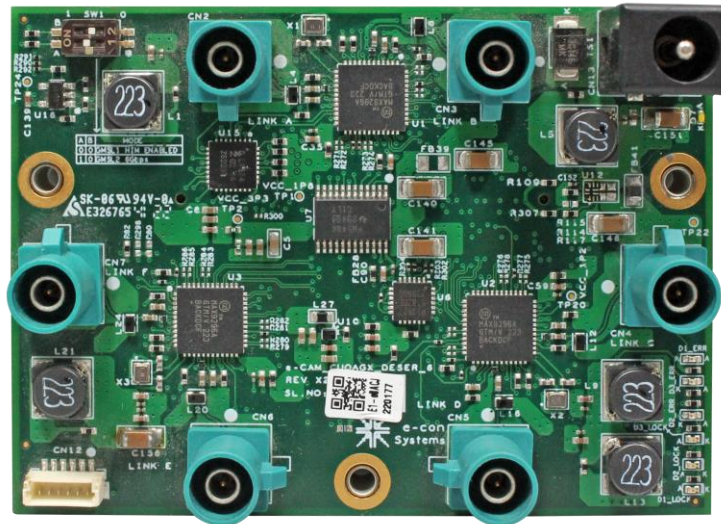


Figure 2: e-CAM_YUV-GMSL-PRODUCTS Deserializer Board

e-CAM_YUV-GMSL-PRODUCTS is a camera having image sensor from onsemi™ along with ISP and the GMSL2 serializer. The serializer in e-CAM_YUV-GMSL-PRODUCTS is based on Maxim GMSL serializer and has all the circuitry required for bidirectional data communication, power, and data separation and so on.

The deserializer board has six FAKRA connectors, through which you can connect the e-CAM_YUV-GMSL-PRODUCTS camera module to any connector. The deserializer board has a power jack connector for providing external power supply, this external power is used for supplying power to the e-CAM_YUV-GMSL-PRODUCTS camera modules over coaxial cable using FAKRA connector.

The dimension of deserializer board is 75 mm x 55 mm. e-CAM_YUV-GMSL-PRODUCTS is available with both 3m and 15m coaxial cable options for power over coax applications.

Board Handling Procedure

This section describes how to connect e-CAM_YUV-GMSL-PRODUCTS to the Jetson AGX Orin™ developer kit. The steps required to set up e-CAM_YUV-GMSL-PRODUCTS board with Jetson AGX Orin™ developer kit are as follows:

- [Connecting Coaxial Cable between Enclosure and Deserializer Board](#)
- [Connecting e-CAM_YUV-GMSL-PRODUCTS with Jetson Orin Developer Kit](#)
- [Powering ON e-CAM_YUV-GMSL-PRODUCTS Camera Modules](#)

Connecting Coaxial Cable between e-CAM_YUV-GMSL-PRODUCTS and Deserializer Board

Deserializer board has an **SW1** switch for selecting GMSL mode. The location of SW1 is shown below.

SW1

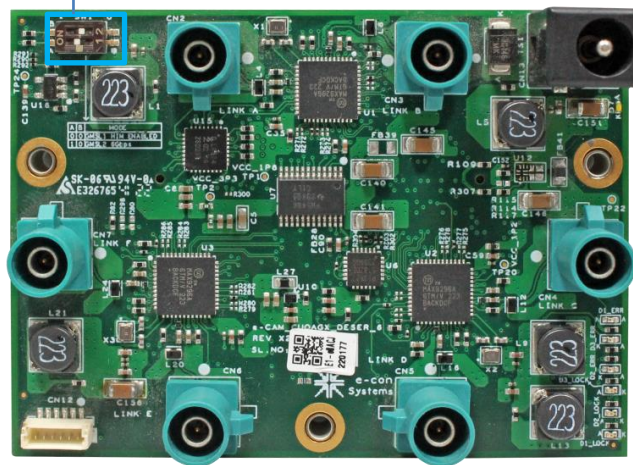


Figure 3: Position of SW1 Switch

The below table describes the different modes in the switch.

Table 3: Modes in Switch

A	B	Mode
0	0	GMSL 1 HIM Enabled
0	1	GMSL 2 3Gbps
1	0	GMSL 2 6Gbps (default)
1	1	GMSL 1 HIM Disabled

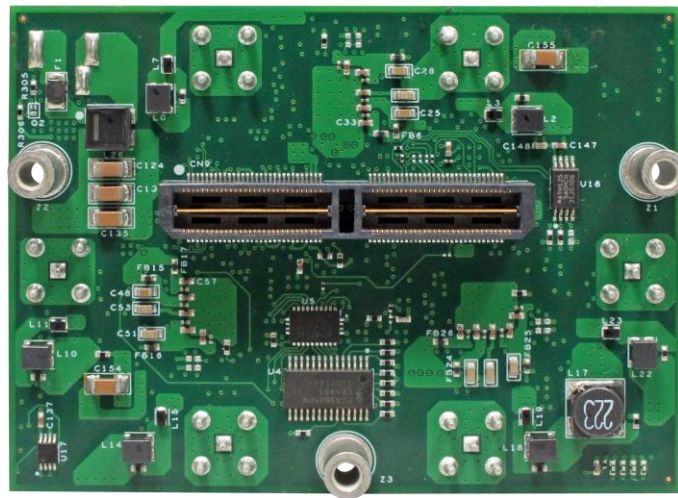
By default, the GMSL2 6Gbps mode is enabled so the position of A and B is 1 and 0 respectively.

To connect coaxial cable between e-CAM_YUV-GMSL-PRODUCTS and deserializer board, follow these steps:

1. Stick the provided bumpers to mounting holes before inserting coaxial cable to deserializer board.

Note: The above step prevents board slanting issues that happens when mated with AGX Orin™ development kit.

The following figure shows the mounting holes location for placing bumper.



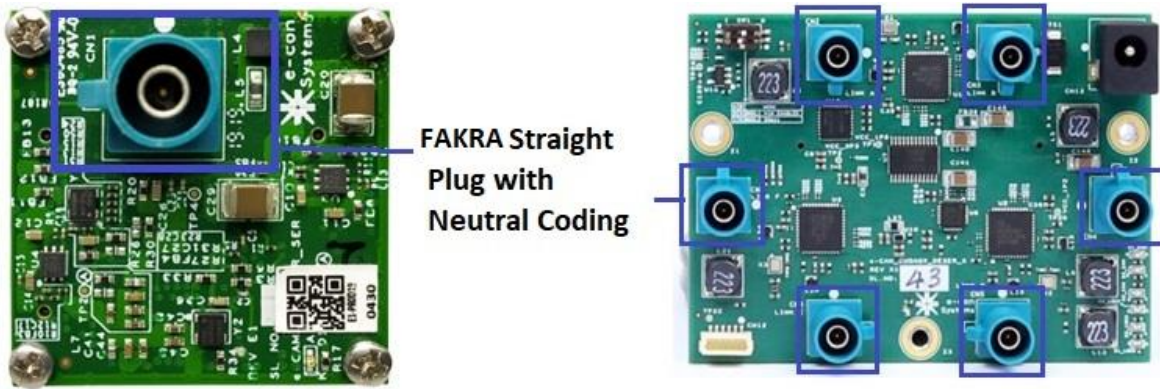


Figure 5: FAKRA Plug on Serializer and Deserializer Board

3. Insert the coaxial cable to FAKRA connector.

The coaxial cable is provided along with the kit. This cable has neutrally coded FAKRA straight jack on either end, to be inserted into the FAKRA plug on the e-CAM_YUV-GMSL-PRODUCTS camera module/deserializer board.

4. Insert the FAKRA jack of the coaxial cable over the plug, such that the jack slides over the mechanical key of the plug.
5. Push the cable such that it locks properly with jack on board.

The following figure shows the properly inserted cable position to the FAKRA plug.

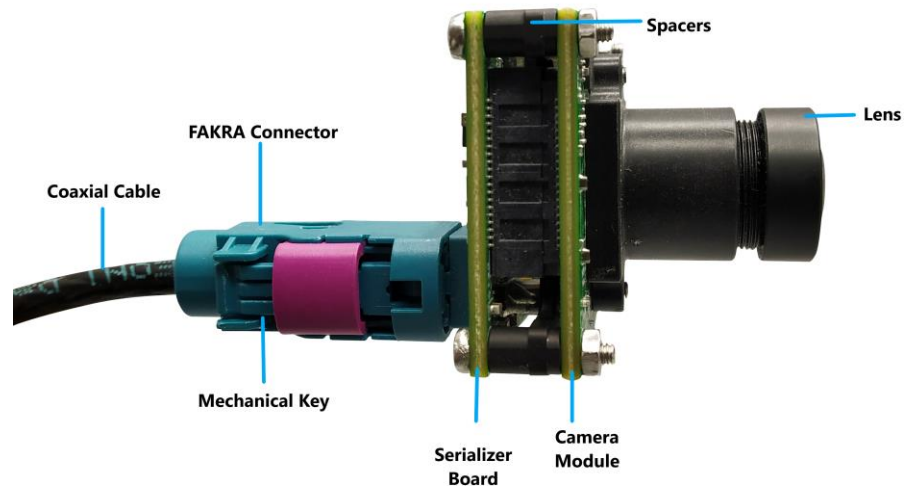


Figure 6: Inserting Coaxial Cable to FAKRA Plug

Note:

- Any of the coaxial cable end can be inserted to the plugs of enclosure and deserializer board.
- For more details in removing coaxial cable from FAKRA plug, refer to *FAQ* section.

The following figure shows the coaxial cable connected to e-CAM_YUV-GMSL-PRODUCTS camera module and deserializer board.

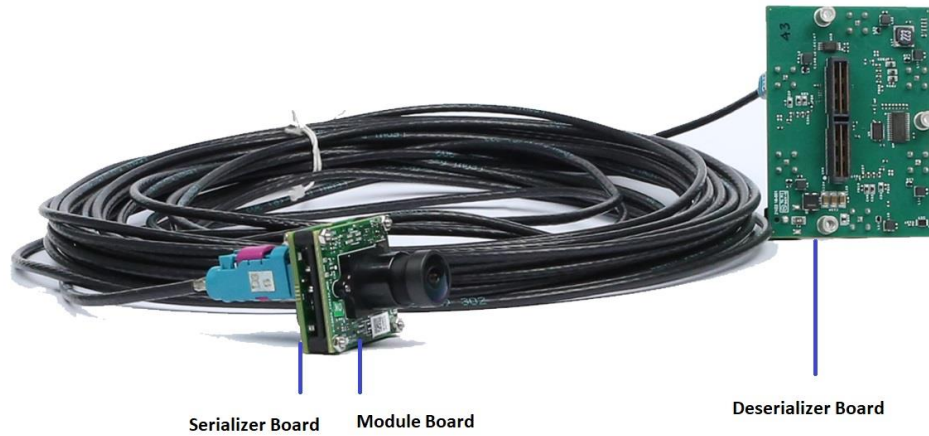


Figure 7: Coaxial cable connected to e-CAM_YUV-GMSL-PRODUCTS (Reference image)

Connecting e-CAM_YUV-GMSL-PRODUCTS with Jetson Orin Developer Kit

The e-CAM_YUV-GMSL-PRODUCTS deserializer board must be mounted to J509 camera interface connector of Orin™ development kit as follows:

1. Insert the CN9.1 pin of e-CAM_YUV-GMSL-PRODUCTS with J509.1 pin of Orin™ development kit.

The e-CAM_YUV-GMSL-PRODUCTS base board location in Jetson AGX Orin™ development kit is shown in following figure.

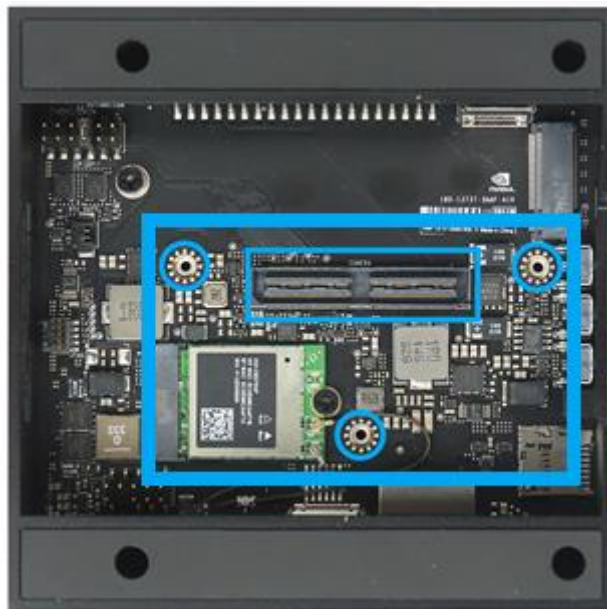


Figure 8: e-CAM_YUV-GMSL-PRODUCTS Base Board Location in Jetson AGX Orin Development Kit

Note: Check the pin 1 marking between deserializer board and Jetson AGX Orin™ kit before mating, refer the below image.

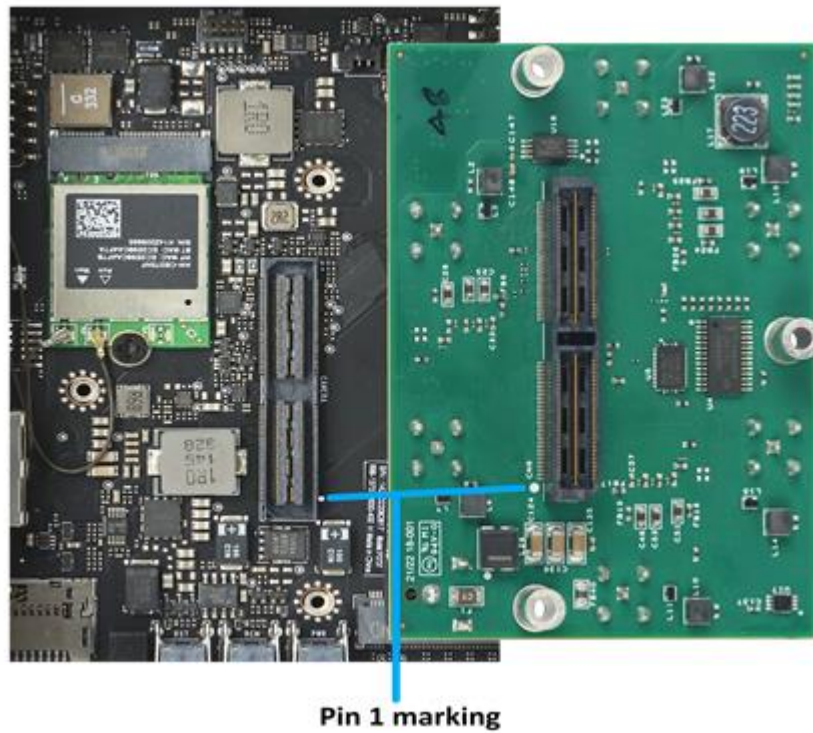


Figure 9: Location of Pins on Deserializer Board and Jetson Orin Development Kit

e-CAM_YUV-GMSL-PRODUCTS deserializer board connected with J509 connector of Orin™ development kit.



Figure 10: Deserializer connected with Jetson AGX Orin™ Development Kit

- Use the provided M2.5 screws for mounting e-CAM_YUV-GMSL-PRODUCTS deserializer board with Orin™ development kit for mechanical stability as shown in Figure 10.

Deserializer board has an SW1 switch for selecting GMSL mode. Select GMSL2 6Gbps mode by selecting “1 0” for A & B position. The location of SW1 is shown below.

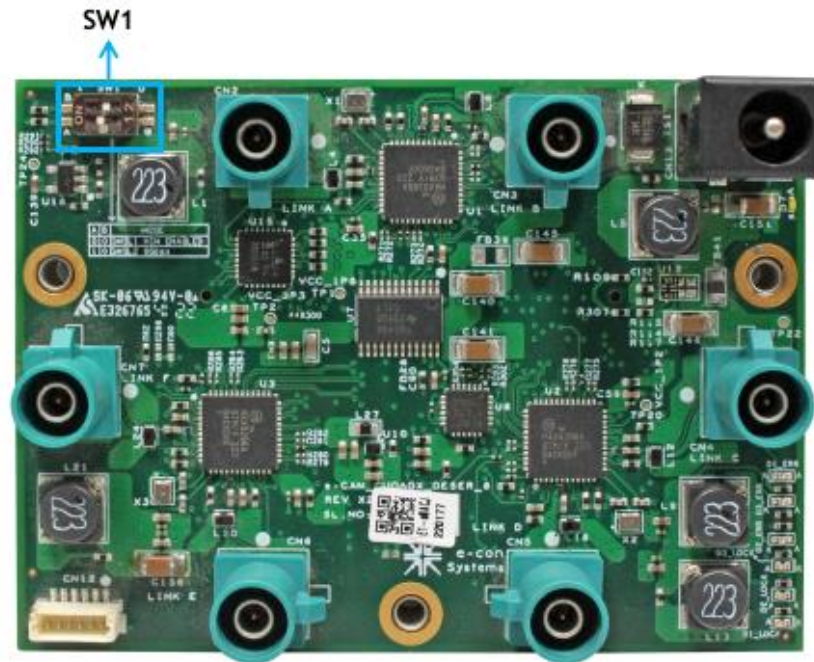


Figure 11: Position of SW1 Switch

The below table describes the different modes in the switch.

A	B	Mode
0	0	GMSL 1 HIM Enabled
0	1	GMSL 2 3Gbps
1	0	GMSL 2 6Gbps (default)
1	1	GMSL 1 HIM Disabled

Table 4: Modes in Switch

- Identify the FAKRA connector on board.

Both the serializer and deserializer boards are provided with high frequency FAKRA straight plug-in water-blue color. The FAKRA connectors are neutrally coded, enabling FAKRA jacks of variable coding to be used in coaxial cable, if required.

The following figure shows the FAKRA plug on e-CAM_YUV-GMSL-PRODUCTS and deserializer boards.

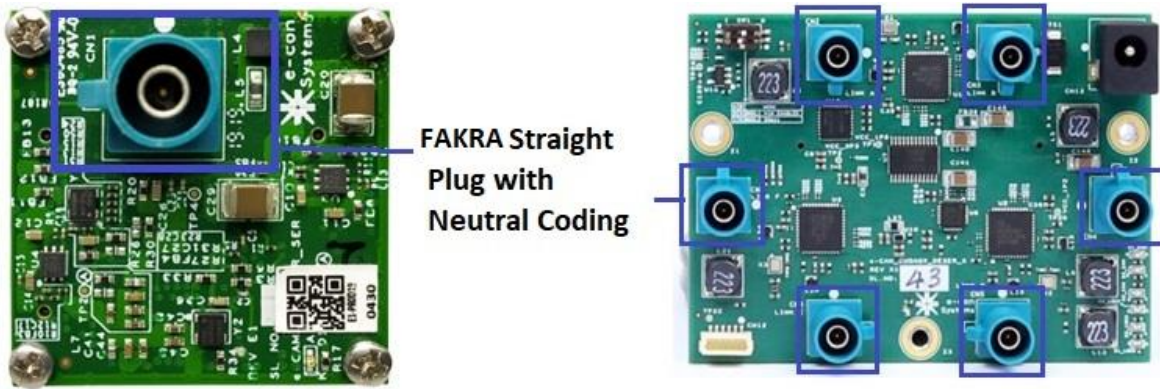


Figure 12: FAKRA Plug on Serializer and Deserializer Boards

4. Insert the coaxial cable to FAKRA connector.

The coaxial cable is provided along with the kit. The cable has a neutrally coded FAKRA straight jack in water-blue color on either end, to be inserted into the FAKRA plug on the serializer/deserializer boards.

5. Insert the FAKRA jack of the coaxial cable over the plug, such that the jack slides over the mechanical key of the plug.
6. Push the cable such that it locks properly with jack on board.

The following figure shows the cable inserted to the FAKRA plug.

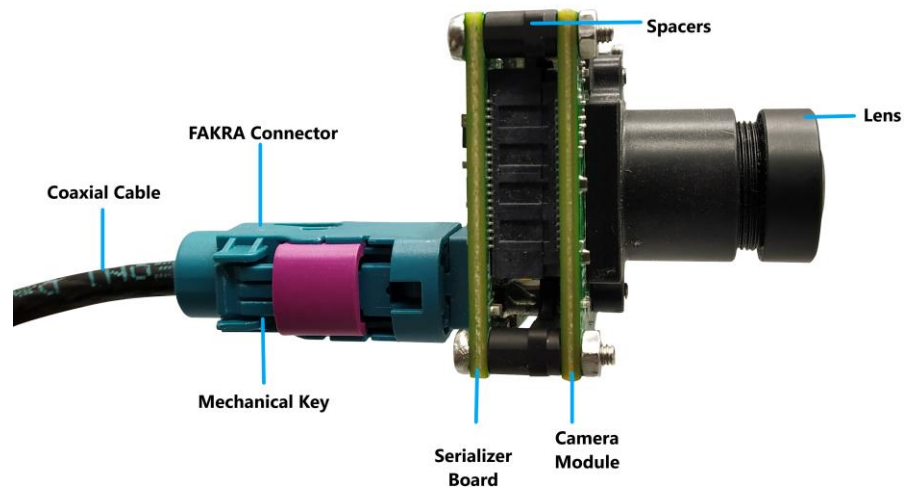


Figure 13: Inserting Coaxial Cable to FAKRA Plug of e-CAM_YUV-GMSL-PRODUCTS Camera Module (Reference image)

Note:

- Any end of the cable can be inserted in any order to the plugs of serializer and deserializer boards.
- For more details in removing coaxial cable from FAKRA plug, refer to *FAQ* section.

The following figure shows the coaxial cable connected to deserializer boards.

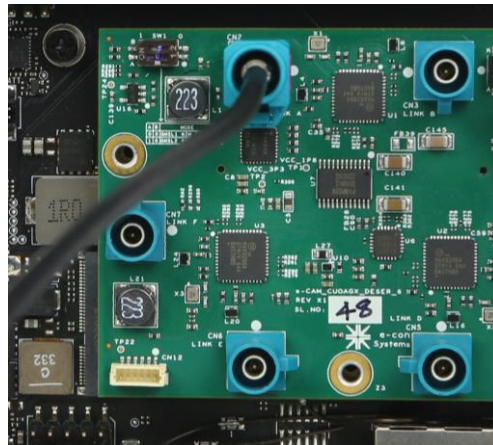


Figure 14: Inserting Coaxial Cable to FAKRA Plugs of e-CAM_YUV-GMSL-PRODUCTS Deserializer

Note: The actual coaxial cable may differ from cable shown in the image.

With e-CAM_YUV-GMSL-PRODUCTS connected to Jetson Orin™ development kit, connect the e-con provided 12V 2A power supply to DC jack of e-CAM_YUV-GMSL-PRODUCTS deserializer board for powering camera modules.

The location of DC jack on e-CAM_YUV-GMSL-PRODUCTS deserializer board is shown in following figure.

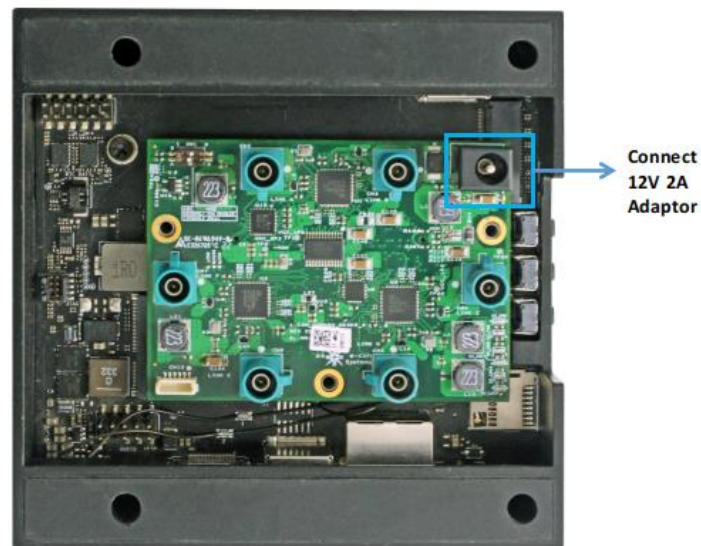


Figure 15: DC Jack Connector location in e-CAM_YUV-GMSL-PRODUCTS Deserializer Board

The power supply connected to DC jack of e-CAM_YUV-GMSL-PRODUCTS deserializer board is shown in following figure.



Figure 16: 12V Power Adapter connected to DC Jack of e-CAM_YUV-GMSL-PRODUCTS Deserializer Board

Powering ON e-CAM_YUV-GMSL-PRODUCTS with Jetson AGX Orin Development Kit

With e-CAM_YUV-GMSL-PRODUCTS connected to Jetson AGX Orin™ development kit, the steps to power up and configure the Orin™ setup are as follows:

1. Connect 20V, 4.5A supply to DC jack to power ON the Jetson Orin™ development kit.

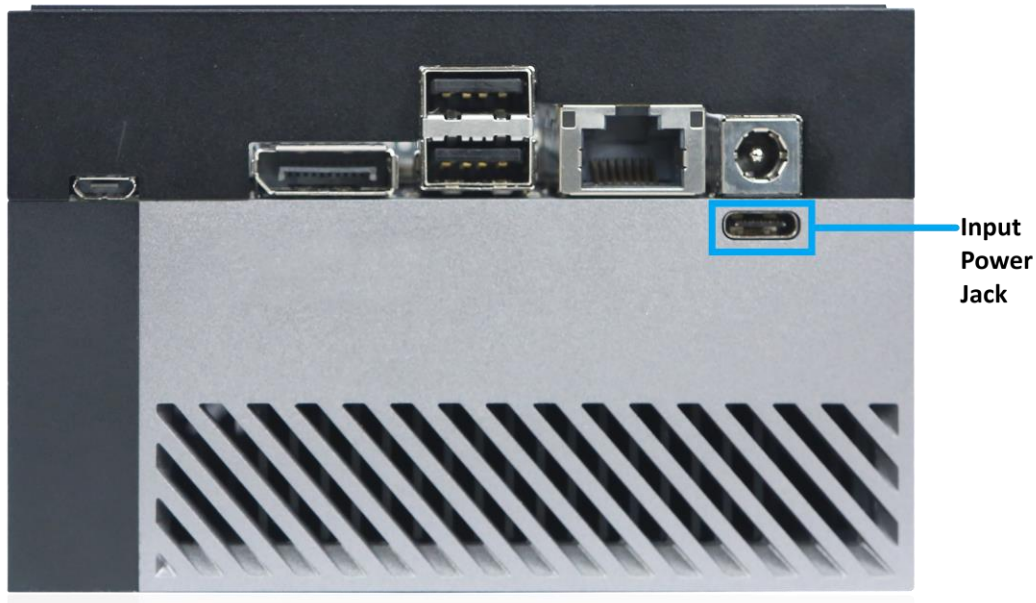


Figure 17: Location of Power Jack on Jetson AGX Orin Development Kit



Figure 18: Power Adaptor connected to Power Jack on Jetson AGX Orin Development Kit

2. Press the power switch of Jetson development kit to boot the device.

The location of power switch on the Jetson AGX Orin™ development kit is shown in following figure.



Figure 19: Location of Power Switch on Jetson AGX Orin Development Kit

After powering ON the Jetson AGX Orin™ development kit, the white color LED on Jetson AGX Orin™ development kit will glow. This serves as an indication for power-up of base board as shown in following figure.

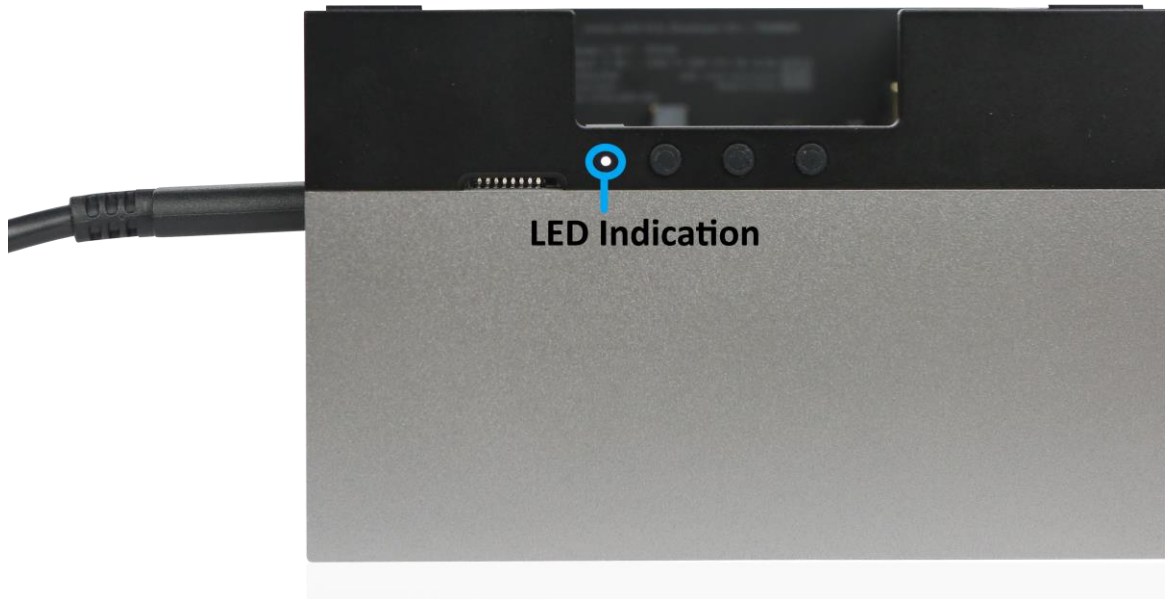


Figure 20: Power ON Indication White LED glowing on Jetson AGX Orin Development Kit

Software Quick Setup

This section describes the extracting and flashing of e-con Systems provided binaries.

The commands and output messages in this manual are represented by different colors as listed in following table.

Table 5: Notation of Colors

Color	Notation
Blue	Commands running in Development PC
Red	Commands running in Jetson Board
Orange	Output messages in the Jetson Board

Before extracting and flashing the e-con provided binaries in Jetson AGX Orin™ development kit, flash the Jetpack 6.0 provided by NVIDIA®, using the SDK manager.

The steps to flash the Jetpack are as follows:

1. Download the SDK manager using this <https://developer.nvidia.com/nvidia-sdk-manager> link.
2. Sign up to an account in NVIDIA® developer site to use the SDK manager.
3. Run the following command to install the SDK Manager in the host PC.

```
sudo apt-get install ./sdkmanager-  
[version].[build#].deb
```

After installing the SDK manager in the host PC, follow the instructions in the link <https://docs.nvidia.com/sdk-manager/install-with-sdgm-jetson/index.html>.

4. Copy the release package into the HOME directory of the flashed Jetson AGX Orin™ development kit.

To extract and flash the binaries, follow these steps:

1. This software package has the following products support.
 1. STURDeCAM20_CUOAGX/NileCAM20_CUOAGX (Product code = 20)
 2. STURDeCAM21_CUOAGX/NileCAM21_CUOAGX (Product code = 21)
 3. STURDeCAM25_CUOAGX/NileCAM25_CUOAGX (Product code = 25)
 4. STURDeCAM81_CUOAGX/NileCAM81_CUOAGX (Product code = 81)

Run the following commands to extract the release package in the Jetson™ development kit to obtain the binaries.

```
tar -xaf e-CAM_YUV-GMSL-
PRODUCTS_JETSON_<L4T_version>_<release_date>_<release_
version>.tar.xz

cd e-CAM_YUV-GMSL-
PRODUCTS_JETSON_<L4T_version>_<release_date>_<release_
version>
```

The package contains the necessary tools to immediately flash the binaries in Jetson AGX Orin™ with the kernel, camera drivers, and application.

3. Run the following commands in the Jetson™ kit to install the binaries.

```
sudo -E ./install_binaries.sh <Product code>
```

(ex. For NileCAM20_CUOAGX -> sudo -E ./install_scripts.sh 20)

Device will reboot automatically once the installation is successful.

The steps to be followed in the development board for launching the application are as follows:

1. Run the following command to check whether all the cameras are initialized properly.

```
dmesg | grep -i 'Detected <sensor name> sensor'
```

The output message appears as shown below.

```
Detected <sensor name> sensor
```

The number of times the above message displayed corresponds to the number of cameras connected to the Jetson™ development kit.

<sensor name> is based on the product name and please refer the table below to find the sensor name:

Product Name	Sensor Name
NileCAM20_CUOAGX/ STURDeCAM20_CUOAGX	ar0230
NileCAM21_CUOAGX/ STURDeCAM21_CUOAGX	ar0233
NileCAM25_CUOAGX	ar0234
NileCAM81_CUOAGX	ar0821

2. Run the following command to check the presence of video nodes.

```
ls /dev/video*
```


The output message appears as shown below.

```
/dev/video*
```

Where (*) denotes the number of cameras connected to the Jetson™ development kit.

3. Run the following command to set the power mode to maximum for better performance.

```
sudo nvpmodel -m 0
```

4. Run the following Jetson clocks command before launching the ecam_tk1_guvcview application in Jetson™ development kit.

```
sudo jetson_clocks
```

5. Run the following command to launch the application.

```
ecam_tk1_guvcview
```

When the application is launched, you can view the screen similar to the screen shown below.

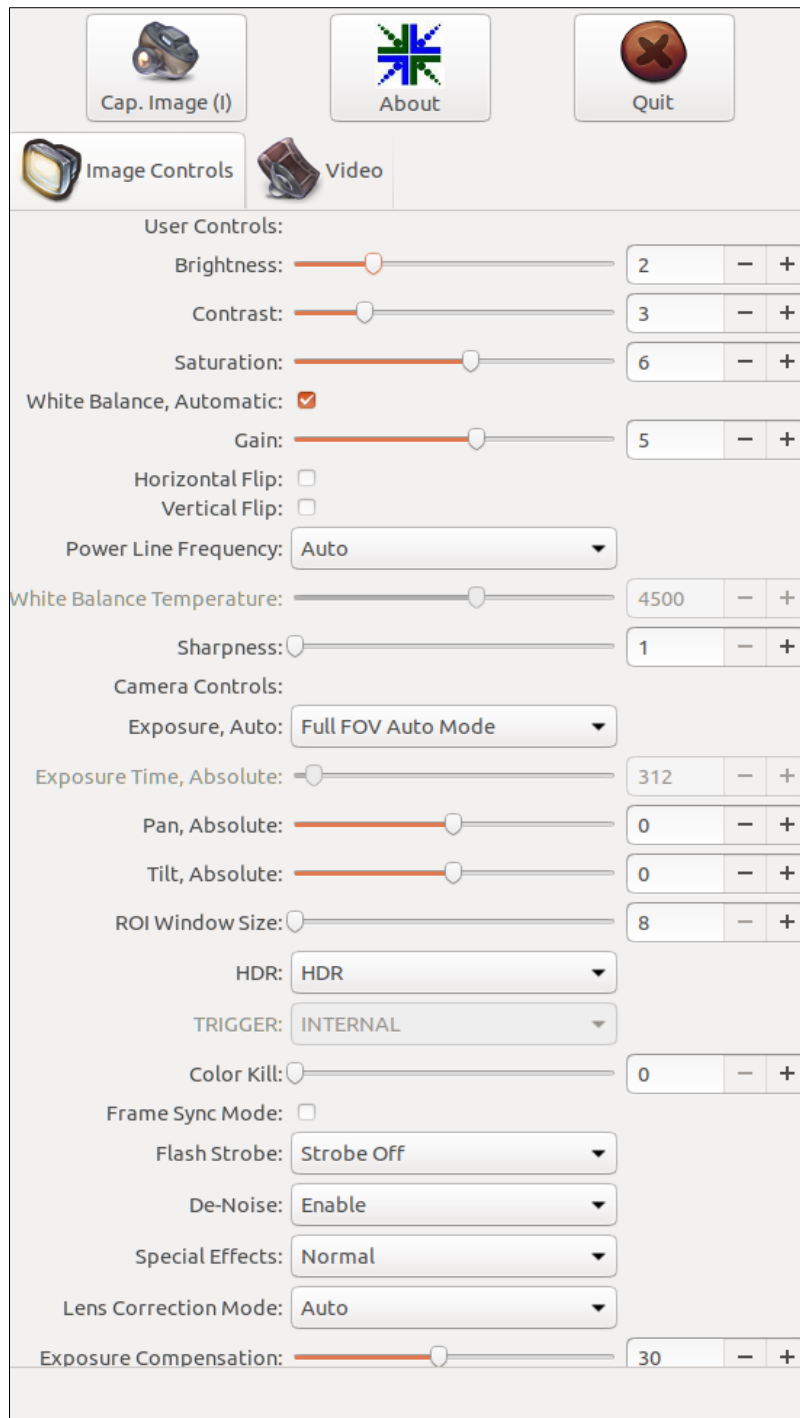


Figure 28: Initial Reference

Window Appearance when Application is launched.

Prerequisites

This section describes the steps for building and installing the kernel.

Setting Up Environment

The steps to setup the environment are as follows:

1. Run the following commands to setup the required environment variables.

```
mkdir -p top_dir
mkdir top_dir/tool_chain
export TOP_DIR=<absolute_path_to>/top_dir
export RELEASE_PACK_DIR=$TOP_DIR/e-CAM_YUV-GMSL-
PRODUCTS_JETSON_<L4T_version>_<release_date>_<release_
version>
export L4T_DIR=$TOP_DIR/Linux_for_Tegra
export LDK_ROOTFS_DIR=$TOP_DIR/Linux_for_Tegra/rootfs
export ARCH=arm64
export TOOL_CHAIN=$TOP_DIR/tool_chain
export CROSS_COMPILE=$TOOL_CHAIN/aarch64--glibc--
stable-2022.08-1/bin/aarch64-buildroot-linux-gnu-
export NVIDIA_SRC=$L4T_DIR/source
export KERNEL_HEADERS=$NVIDIA_SRC/kernel/kernel-jammy-
src
export INSTALL_MOD_PATH=$LDK_ROOTFS_DIR
export TEGRA_KERNEL_OUT=$NVIDIA_SRC/out/nvidia-linux-
header
export SENSOR_DRIVER=e-CAM_YUV_GMSL
```

2. Run the following command to copy the release package tar file to the staging directory.

```
mv <location_of>/e-CAM_YUV-GMSL-
PRODUCTS_JETSON_AGX_ORIN_<L4T_version>_<release_date>_
<release_version>.tar.xz $TOP_DIR
```

Note: The above steps must be performed in a single terminal such that the environment variables are preserved.

Downloading Requirements

For building the kernel, a cross compiler toolchain and other tools necessary for compiling are required.

The steps to download the requirements for building the kernel are as follows:

1. Download the required toolchain from NVIDIA® website using the link listed in the below table.

Table 6: Required Toolchain

S.NO	Title	Version	Download Link
1	GCC Toolchain	11.3.0	https://developer.nvidia.com/downloads/embedded/l4t/r36_release_v3.0/toolchain/aarch64--glibc--stable-2022.08-1.tar.bz2

- a) Run the following commands to extract the package in Host PC.

```
mv ~/Downloads/aarch64--glibc--stable-2022.08-1.tar.bz2 $TOP_DIR/tool_chain
cd $TOP_DIR/tool_chain
tar -xf aarch64--glibc--stable-2022.08-1.tar.bz2
```

- b) Run the following command to add CROSS_COMPILER to PATH environment variable for building kernel source.

```
export PATH=$TOOL_CHAIN/bin:${PATH}
```

2. Run the following commands to download the required package for extracting sources.

```
sudo apt-get update
sudo apt-get install qemu-user-static
sudo apt-get install build-essential
sudo apt-get install bc
sudo apt-get install lbzip2
sudo apt-get install python
sudo apt-get install flex
sudo apt-get install openssl
sudo apt-get install libssl-dev
```

3. Download the required L4T release package and sample rootfs from NVIDIA® website using <https://developer.nvidia.com/embedded/downloads> link.

The steps to download and copy the package to staging directory are as follows:

- a) Download the packages from the NVIDIA® website as listed in below table.

Table 7: Packages for Jetson Orin NX

S.NO	Title	Version	Download Link
1	L4T Jetson Orin Nano™ + Jetson Orin Nano™ Driver Package	36.3.0	https://developer.nvidia.com/downloads/embedded/l4t/r36_release_v3.0/release/jetson_linux_r36.3.0_aarch64.tbz2
2	L4T Jetson Orin Nano™ + Jetson Orin Nano™ Sample Rootfs	36.3.0	https://developer.nvidia.com/downloads/embedded/l4t/r36_release_v3.0/release/tegra_linux_sample-root-filesystem_r36.3.0_aarch64.tbz2

- b) Run the following commands to copy the downloaded package to staging directory.

```
cp $HOME/Downloads/Jetson_Linux_R36.3.0_aarch64.tbz2 $TOP_DIR
cp $HOME/Downloads/Tegra_Linux_Sample-Root-Filesystem_R36.3.0_aarch64.tbz2 $TOP_DIR
```

Extracting and Preparing L4T

The steps for extracting and preparing L4T must be performed in host PC as follows:

1. Run the following commands to extract the downloaded L4T release package to navigate a folder with the name Linux_for_Tegra.

```
cd $TOP_DIR
tar xf Jetson_Linux_R36.3.0_aarch64.tbz2
```

2. Run the following commands to extract the sample rootfs to the rootfs directory which is present inside the Linux_for_Tegra directory.

```
sudo tar xpf Tegra_Linux_Sample-Root-Filesystem_R36.3.0_aarch64.tbz2 -C Linux_for_Tegra/rootfs/
```

3. Run the following commands to set the package to be ready to flash binaries.

```
cd $L4T_DIR
sudo ./apply_binaries.sh
sudo ./tools/l4t_flash_prerequisites.sh
```

Extracting Release Package

Run the following commands to extract the e-CAM_YUV-GMSL-PRODUCTS release package.

```
cd $TOP_DIR
```



```
tar -xaf e-CAM_YUV-GMSL-  
PRODUCTS_JETSON_AGX_ORIN_<L4T_version>_<release_date>_<re  
lease_version>.tar.xz
```

To know more about the release package, please refer to the *e-CAM_YUV_GMSL_Release_Package_Manifest_Rev_<ver>.pdf*.

Installation Procedure

Building from Source

You can use the patch provided by e-con Systems to use your own kernel image binary and modules along with the e-CAM_YUV-GMSL-PRODUCTS camera on the Jetson AGX Orin™ development kit.

Downloading and Configuring the Kernel

This section describes how you can download and configure the kernel for Jetson AGX Orin™ development kit.

Download the kernel source code for L4T from the NVIDIA® website using <https://developer.nvidia.com/embedded/downloads> link.

The steps to download and configure the kernel for Jetson AGX Orin™ development kit are as follows:

1. Download the packages from the NVIDIA® website as listed in below table.

Table 8: Packages for Jetson AGX Orin™

Title	Version	Download Link
L4T Jetson AGX Orin™ R36.3.0 Sources	36.3.0	https://developer.nvidia.com/downloads/embedded/l4t/r36_release_v3.0/sources/public_sources.tbz2

2. Run the following command to copy the downloaded file to staging directory.

```
cp $HOME/Downloads/public_sources.tbz2 $TOP_DIR/
```

3. Run the following commands to extract the downloaded kernel source code to any path on the host Linux PC.

```
tar -xjpf public_sources.tbz2  
cd $NVIDIA_SRC
```

4. Run the following command to extract the kernel source code.

```
tar xf kernel_src.tbz2  
tar xf kernel_oot_modules_src.tbz2  
tar xf nvidia_kernel_display_driver_source.tbz2
```

5. Run the following commands to make sure that all the four patch commands are applied properly in the kernel source. Run the following commands to make

sure that all the three patch commands are applied properly in the kernel source.

```
patch -p1 -i $RELEASE_PACK_DIR/Kernel/e-CAM-
YUV_GMSL_JETSON_ORIN_<L4T_version>_kernel.patch --dry-
run

patch -p1 -i $RELEASE_PACK_DIR/Kernel/e-CAM-
YUV_GMSL_JETSON_ORIN_<L4T_version>_oot.patch --dry-run

patch -p1 -i $RELEASE_PACK_DIR/Kernel/e-CAM-
YUV_GMSL_JETSON_ORIN_<L4T_version>_dtb.patch --dry-run

mkdir -p sensor_driver

patch -d sensor_driver -p1 -i
$RELEASE_PACK_DIR/Kernel/e-CAM-
YUV_GMSL_JETSON_ORIN_<L4T_version>_module.patch --dry-
run
```

Note: All four patches are for kernel, nvidia-oot, device-tree, and camera-module driver.

4. Run the following commands to apply the patch file to the kernel source code if there is no error from dry-run command.

```
patch -p1 -i $RELEASE_PACK_DIR/Kernel/e-CAM-
YUV_GMSL_JETSON_ORIN_<L4T_version>_kernel.patch

patch -p1 -i $RELEASE_PACK_DIR/Kernel/e-CAM-
YUV_GMSL_JETSON_ORIN_<L4T_version>_oot.patch

patch -p1 -i $RELEASE_PACK_DIR/Kernel/e-CAM-
YUV_GMSL_JETSON_ORIN_<L4T_version>_dtb.patch

patch -d sensor_driver -p1 -i
$RELEASE_PACK_DIR/Kernel/e-CAM-
YUV_GMSL_JETSON_ORIN_<L4T_version>_module.patch
```

Building and Installing the Kernel

1. Run the following commands to build and install the kernel image and modules to support the Jetson AGX Orin™ development kit.

```
make -C kernel
sudo -E make install -C kernel
make modules
sudo -E make modules_install
make dtbs
```

2. Run the following commands to copy the kernel Image to Linux_for_Tegra (L4T_DIR) flashing path.

```
sudo cp $NVIDIA_SRC/kernel/kernel-jammy-
src/arch/arm64/boot/Image $L4T_DIR/kernel/Image -f
```

3. Run the following commands to copy the dtb to Linux_for_Tegra (L4T_DIR) flashing path.

For Jetson AGX Orin™ 32 GB Development Kit,

```
sudo cp $NVIDIA_SRC/nvidia-oot/device-  
tree/platform/generic-dts/dtbs/tegra234-p3737-0000+p3701-  
0000-nv.dtb $L4T_DIR/kernel/dtb/tegra234-p3737-  
0000+p3701-0000-nv.dtb -f
```

For Jetson AGX Orin™ 64 GB Development Kit,

```
sudo cp $NVIDIA_SRC/nvidia-oot/device-  
tree/platform/generic-dts/dtbs/tegra234-p3737-0000+p3701-  
0005-nv.dtb $L4T_DIR/kernel/dtb/tegra234-p3737-  
0000+p3701-0005-nv.dtb -f
```

4. Run the following commands to copy the dtbo file to Linux_for_Tegra (L4T_DIR) flashing path.

For STURDeCAM20/ NileCAM20,

```
sudo cp $NVIDIA_SRC/nvidia-oot/device-  
tree/platform/generic-dts/dtbs/tegra234-p3737-camera-  
overlay_ar0230_two_lane.dtbo $LDK_ROOTFS_DIR/boot/
```

For STURDeCAM21/ NileCAM21,

```
sudo cp $NVIDIA_SRC/nvidia-oot/device-  
tree/platform/generic-dts/dtbs/tegra234-p3737-camera-  
overlay_ar0233_two_lane.dtbo $LDK_ROOTFS_DIR/boot/
```

For NileCAM25,

```
sudo cp $NVIDIA_SRC/nvidia-oot/device-  
tree/platform/generic-dts/dtbs/tegra234-p3737-camera-  
overlay_ar0234_two_lane.dtbo $LDK_ROOTFS_DIR/boot/
```

For NileCAM81,

```
sudo cp $NVIDIA_SRC/nvidia-oot/device-  
tree/platform/generic-dts/dtbs/tegra234-p3737-camera-  
overlay_ar0821_two_lane.dtbo $LDK_ROOTFS_DIR/boot/
```

Note: Even if the image is custom built, the kernel configuration must have module versioning support for the camera driver to work.

Modifying the Rootfs

Run the following command to modify additional files in the rootfs for the proper functioning of the e-CAM_YUV-GMSL-PRODUCTS camera on the Jetson AGX Orin™ development kit.

```
sudo cp $RELEASE_PACK_DIR/misc/v4l2-compliance  
$LDK_ROOTFS_DIR/usr/local/bin/ -f  
sudo chmod +x $LDK_ROOTFS_DIR/usr/local/bin/v4l2-  
compliance
```

Copying the MCU firmware to Rootfs

Run the following command to copy the mcu firmware files to the rootfs for the proper functioning of the e-CAM_YUV-GMSL-PRODUCTS camera on the Jetson AGX Orin™ development kit.

For Jetson AGX Orin™,

```
sudo cp $RELEASE_PACK_DIR/firmware/*  
$LDK_ROOTFS_DIR/usr/lib/firmware/
```

Flashing Jetson Development Kit

The steps to flash the Jetson™ development kit are as follows:

1. Connect the USB Type-C cable to the host PC and the USB Type-C port of Jetson AGX Orin™ development kit.



Figure 30: The location of USB-C port on the Jetson Orin™ development kit is shown in above figure.

2. Set the board to recovery mode, as mentioned in below steps:
 - a. Press and hold the **Recovery** button of Jetson AGX Orin™ development kit.
 - b. Press the **Power** button of Jetson AGX Orin™ development kit.
 - c. Release **Power** button and then Release **Recovery** button.

The location of Recovery and Power buttons on the Jetson AGX Orin™ development kit is shown in below figure.



Figure 32: Location of Recovery and Power Buttons on Jetson AGX Orin Development Kit

If the board is successfully changed to recovery mode, the Jetson AGX Orin™ development kit will be enumerated as an USB device to the host PC.

3. Run the following command to verify whether the board is in recovery mode.

```
lsusb
```

The output message appears for Jetson AGX Orin™ as shown below.

```
Bus 001 Device 102: ID 0955:7020 Nvidia Corp
```

4. Run the following command to flash the Jetson™ development kit from your host PC.

```
cd $L4T_DIR
```

For Jetson AGX Orin™,

```
sudo ./flash.sh jetson-agx-orin-devkit internal
```

Note: Now, the entire eMMC on the Jetson AGX Orin™ development kit and any files present on the device will be erased. It will take about 10-30 minutes to complete, based on the host PC configuration.

5. Configure the Jetson AGX Orin™ device once after complete flashing.

Loading Drivers

This section describes how to load the drivers, install the sample application, and use the sample application with e-CAM_YUV-GMSL-PRODUCTS.

The module drivers for e-CAM_YUV-GMSL-PRODUCTS will be loaded automatically in the Jetson Orin™ development kit during booting.

The steps to load the drivers are as follows:

1. Run the following command to select e-CAM_YUV-GMSL-PRODUCTS device tree overlay file for 2 lane.

For STURDeCAM20/ NileCAM20

```
sudo /opt/nvidia/jetson-io/config-by-hardware.py -n  
2="Jetson Camera STURDeCAM20-AR0230"
```

the output message appears as shown below,

```
Modified /boot/extlinux/extlinux.conf to add following  
DTBO entries:  
/boot/tegra234-p3737-camera-  
overlay_ar0230_two_lane.dtbo  
Reboot system to reconfigure.
```

For STURDeCAM21/ NileCAM21,

```
sudo /opt/nvidia/jetson-io/config-by-hardware.py -n  
2="Jetson Camera STURDeCAM21-AR0233"
```

the output message appears as shown below,

```
Modified /boot/extlinux/extlinux.conf to add following  
DTBO entries:  
/boot/tegra234-p3737-camera-  
overlay_ar0233_two_lane.dtbo  
Reboot system to reconfigure.
```

For NileCAM25,

```
sudo /opt/nvidia/jetson-io/config-by-hardware.py -n  
2="Jetson Camera NileCAM25-AR0234"
```

the output message appears as shown below,

```
Modified /boot/extlinux/extlinux.conf to add following  
DTBO entries:
```

```
/boot/tegra234-p3737-camera-  
overlay_ar0234_two_lane.dtbo  
Reboot system to reconfigure.
```

For NileCAM81,

```
sudo /opt/nvidia/jetson-io/config-by-hardware.py -n  
2="Jetson Camera NileCAM81-AR0821"
```

the output message appears as shown below,

```
Modified /boot/extlinux/extlinux.conf to add following  
DTBO entries:  
/boot/tegra234-p3737-camera-  
overlay_ar0821_two_lane.dtbo  
Reboot system to reconfigure.
```

2. Run the following command to check whether the camera is initialized.

```
sudo dmesg | grep -i "ecam_gmsl"
```

The output message appears as shown below for Jetson AGX Orin™ development kit.

```
subdev ecam_gmsl 9-0043 bound
```

3. Run the following command to check the presence of video node.

```
ls /dev/video*
```

The output message appears as shown below.

```
/dev/video0
```

where (*) denotes the number of cameras connected to the Jetson™ development kit.

The number of video nodes and the number of cameras must be equal. The output message must list one video node if no other cameras are connected to the Jetson AGX Orin™ development kit. This video node can be utilized by any V4L2 application for viewing the camera preview.

e-CAM_TK1-GUVCView Installation

Installing Build Dependencies

The steps to install the build dependencies are as follows:

1. Run the following command in Jetson AGX Orin™ board to enable all the repositories which are required for installing the dependencies.

```
$ sudo apt-add-repository universe
$ sudo apt-get update
$ sudo apt-get install make
```

Note: Make sure that you have connected Jetson AGX Orin™ board to a stable network.

2. Run the following command to install the dependencies in Jetson AGX Orin™ board.

```
$ sudo apt-get install intltool libsdl1.2-dev libusb-
1.0-0-dev libudev-dev libportaudio-ocaml-dev
libpango1.0-dev libatk1.0-dev libgdk-pixbuf2.0-dev
libatk-bridge2.0-dev libcairo2-dev libgtk-3-dev
libavutil-dev libavcodec-dev libzip2
```

Note: If the installation stops with the **Unmet dependencies** error, please run the following command to solve the issue.

```
$ sudo apt --fix-broken install
```

3. Run the following commands to install the FFmpeg dependency package.

```
$ wget -c http://ffmpeg.org/releases/ffmpeg-
2.8.15.tar.xz
$ tar -xf ffmpeg-2.8.15.tar.xz
$ cd ffmpeg-2.8.15
#For fast install change the power mode to maximum
$ sudo nvpmodel -m 0
$ sudo jetson_clocks
$ ./configure
$ make -j4
$ sudo make install
```

Note: You can also use the FFmpeg package source provided in e-CAM_YUV-GMSL-PRODUCTS_<L4T_version>_<release_date>_<release_version>.tar.xz release package at **Application/e-CAM_TK1_guvcview/Source/ecam_tk1_guvcview_dependency.tar.gz** location.

Identifying the Deliverables

The steps for identifying the deliverables are as follows:

1. Copy the release package tar file to the home directory of the board.
2. Run the following commands to extract the e-CAM_YUV-GMSL-PRODUCTS release package.

```
tar -xaf e-CAM_YUV-GMSL-  
PRODUCTS_JETSON_<L4T_version>_<release_date>_<release_  
version>.tar.xz  
  
cd e-CAM_YUV-GMSL-  
PRODUCTS_JETSON_<L4T_version>_<release_date>_<release_  
version>
```

Note: Do not right click to extract the package.

The source code for the ecam_tk1_guvcview application is present in the release package at the following location. **Application/e-CAM_TK1_guvcview/Source/e-CAM_TK1_guvcview-src-1.7.2-<version>.tar.gz.**

Building and Installing e-CAM_TK1-GUVCView from source

The steps to build and install ecam_tk1_guvcview from the source are as follows:

1. Run the following commands to navigate to the application source directory in Jetson AGX Orin™ board and extract the package.

```
cd Application/ecam_tk1_guvcview/Source  
tar -xf e-CAM_TK1_guvcview-src-1.7.2.tar.gz  
cd e-CAM_TK1_guvcview-src-1.7.2/
```

2. Run the following commands to configure the ecam_tk1_guvcview source.

```
./configure --prefix=/usr/local/ecam_tk1  
echo 'export PATH=$PATH:/usr/local/ecam_tk1/bin' >>  
$HOME/.bashrc  
source ~/.bashrc
```

The configuration is required to check the presence of dependent packages and to create the makefiles. If you want to change the **default /usr/local/bin** binary installation directory, add **--prefix = <path_to_install>**.

To know more information on configuring ecam_tk1_guvcview source, run the following command.

```
./configure --help
```

Note: If configuration fails, please contact the Ubuntu's [Help Center](#) to know on how to install any required packages.

3. Run the following make command to build the `ecam_tk1_guvcview` application from source.

```
make -j4
```

4. Run the following make install command to install the built application.

```
sudo make install
```

The `ecam_tk1_guvcview` application will be installed in `/usr/local/ecam_tk1/bin` location of Jetson AGX Orin™ board. This application is used to capture and view video from the camera.

e-multicam Installation

Identifying the Deliverables

This section describes about identifying the deliverables.

The release package contains the application source code and prebuilt application. The commands and output messages in this manual are represented by different colors as shown in below table.

Table 9: Notation of Colors

Color	Notation
Blue	Commands running in development PC
Red	Output message in development PC
Green	Commands running in development board

The steps for identifying the deliverables are as follows:

1. Copy the release package tar file to the home directory of the board.
2. Run the following commands to extract the STURDeCAM20_CUOAGX release package.

```
tar -xaf e-CAM_YUV-GMSL-PRODUCTS_JETSON
_<L4T_version>_<release_date>_<release_version>.tar.xz
cd e-CAM_YUV-GMSL-PRODUCTS_JETSON
_<L4T_version>_<release_date>_<release_version>
```

Note: Do not right click to extract the package.

The source code for the e-multicam.elf application is present in the release package at the following location.

Application/e-multicam/Source/e-multicam.tar.gz

Note: Please refer to *e-multicam* section to build application from the source and refer to **Error! Reference source not found.** section for using the application.

Building and Installing e-multicam from source

This section describes about building and installing the e-multicam.elf application from the source.

The steps to build and install e-multicam.elf application from the source are as follows:

1. Run the following commands to install the dependency Gstreamer-1.0 libraries.

```
$ sudo apt-get update
$ sudo apt-get install libgstreamer-plugins-base1.0-dev libgstreamer1.0-dev libjpeg-dev wmctrl make
```

Note: If the installation process stops with the **could not get lock /var/lib/dpkg/lock** message, run the following command to remove the file and proceed with the installation.

```
$ sudo rm /var/lib/dpkg/lock
```

Note: If the installation stops with the **Unmet dependencies** error, please run the following command to solve the issue.

```
$ sudo apt --fix-broken install
```

2. Run the following commands to enter the source directory.

```
$ cd Application/e-multicam/Source/
$ tar -xf e-multicam.tar.gz
$ cd e-multicam/
```

3. Run the following make command to build the e-multicam application from the source.

```
$ make
```

4. Run the following make install command to install the build application.

```
$ sudo make install
```

The application will be installed in **/usr/local/bin** location.

Reference Documents

This section describes the software and hardware documents of e-CAM_YUV-GMSL-PRODUCTS. You can download the software and hardware documents from [Developer Resources](#) website.

Software Documents

The software documents and its description are listed in following table.

Table 10: Description of Software Documents

S.NO	What I need	Documents to Refer
1	View the single camera stream and change the camera controls using GUI.	e-CAM_YUV-GMSL_Linux_App_User_Manual_<REV>.pdf
2	Use Gstreamer to control the e-CAM_YUV-GMSL-PRODUCTS camera on the Jetson™ development kit.	e-CAM_YUV-GMSL_Gstreamer_Usage_Guide_<REV>.pdf
3	Stream and view all the cameras of e-CAM_YUV-GMSL-PRODUCTS in synchronous mode.	e-CAM_YUV-GMSL_e-multicam_App_User_Manual_<REV>.pdf
4	Information about the directory structure and contents of the release package for e-CAM_YUV-GMSL-PRODUCTS.	e-CAM_YUV-GMSL_Release_Package_Manifest_<REV>.pdf
5	Integrate e-con's e-CAM_YUV-GMSL-PRODUCTS with any host system other than what is supported by e-con Systems directly.	e-CAM217_CUMI0234_MCU_Protocol_App Note_<REV>.pdf/ e-CAM21_CUMI290_MCU_Protocol_AppNote_<REV>.pdf/ e-CAM217_CUMI0234_MCU_Protocol_App Note_<REV>.pdf/ e-CAM82_CUMI0821_MCU_Protocol_AppNote_<REV>.pdf

Hardware Documents

The hardware documents and its description are listed in following table.

Table 11: Description of Hardware Documents

S.NO	Documents Name	Description
1	e-CAM_YUV-GMSL-PRODUCTS_Datasheet_<REV>.pdf	Describes the features, connector pin-out details and mechanical dimensions of e-CAM_YUV-GMSL-PRODUCTS.
2	e-CAM_YUV-GMSL-PRODUCTS_Camera_Module_Datasheet_<REV>.pdf	Describes the features, connector pin-out details and mechanical dimensions of e-CAM_YUV-GMSL-PRODUCTS.
3	e-CAM_YUV-GMSL-PRODUCTS_Lens_Datasheet_<REV>.pdf	Describes the optical specification of lenses used in e-CAM_YUV-GMSL-PRODUCTS
4	e-CAM_YUV-GMSL-PRODUCTS_Mounting_Procedure_<REV>.pdf	Mounting steps for e-CAM_YUV-GMSL-PRODUCTS modules.
5	e-CAM_YUV-GMSL-PRODUCTS_COMBINED_3D.stp	3D drawing of enclosure.

Troubleshooting

In this section, you can view the list of commonly occurring issues and their troubleshooting steps.

The systems are running in low-graphics mode error log appears on screen. How to solve this issue?

This error may occur in Jetson AGX Orin™ when complete package is flashed. To solve this issue, reset the board.

Why the camera is not detected, even if the device is connected and powered ON?

This may occur on the deserializer board when it is not powered ON. Make sure that the deserializer board is powered ON, before the Jetson™ development kit.

Why power ON indication LED of serializer is OFF, even if the device is connected?

When the power indicator LED of Serializer (D2) is OFF (with status LED of base board is ON), it means that power is not applied from base board. Check if the coaxial cable is inserted properly on either end.

Why error LED of deserializer is ON, if device is connected?

Apart from Lock LED, the deserializer board has a RED error LED (D1). The LED glows, when the error in the GMSL data exceeds the set threshold. The glowing of RED LED does not affect the camera streaming.

The following figure shows the Error LED (RED) on Deserializer board.



FAQ

- 1. Does e-CAM_YUV-GMSL-PRODUCTS camera work with any coaxial GMSL cable available in the market?**

Currently, e-con Systems provide a very flexible and robust 3m/15m coaxial GMSL cable along with this kit. For more lengths and customizations, please write to camerasolutions@e-consystems.com with your requirement.

- 2. Does e-CAM_YUV-GMSL-PRODUCTS camera support NVIDIA® TX2i?**

Currently, this camera supports NVIDIA® Orin™ AGX only. For customization, please write to camerasolutions@e-consystems.com with your requirement.

- 3. Does e-CAM_YUV-GMSL-PRODUCTS camera support OpenCV?**

e-CAM_YUV-GMSL-PRODUCTS works using V4L2 APIs and is V4L2 compliant. So, any V4L2 based application can be used to access this camera. OpenCV is also compatible since it uses V4L2 to access the camera.

Please refer to https://www.e-consystems.com/Articles/Camera/accessing_cameras_in_opencv_with_high_performance.asp for detailed information about OpenCV support in e-con Systems cameras.

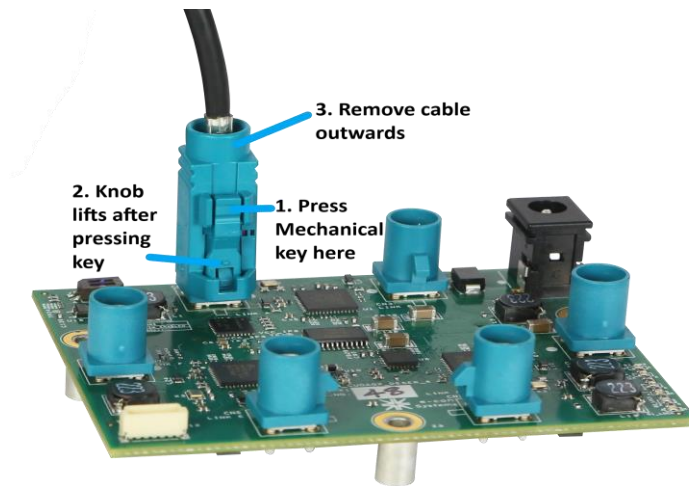
- 4. How can I get the updated package?**

Please login to the [Developer Resources](#) website and download the latest release package.

- 5. How to remove the coaxial cable from FAKRA plug?**

Care must be taken while removing the coaxial cable from the PCB. The FAKRA jack on coaxial cable locks with mechanical key of the FAKRA plug on board. The lock between plug and jack needs to be released, to remove the cable. On pressing the jack at the specified location below, the knob lifts and the cable can be removed.

The following figure shows the stages of removing FAKRA jack (coaxial cable) from PCB.



What's Next?

After understanding the specifications of camera daughter board and instructions on how to use this daughter board with Jetson AGX Orin™ development kit, you can refer to the following documents to understand more about e-CAM_YUV-GMSL-PRODUCTS.

- *Respective product's Linux App User Manual*

Glossary

API: Application Program Interface.

CMOS: Complementary Metal Oxide Semiconductor.

CSI: Camera Serial Interface.

GMSL: Gigabit Multimedia Serial Link.

GUI: Graphical User Interface.

HDMI: High Definition Multimedia Interface.

ISP: Image Signal Processor.

LED: Light-Emitting Diode.

MCU: Micro Controller Unit.

MIPI: Mobile Industry Processor Interface.

PCB: Printed Circuit Board.

USB: Universal Serial Bus.

V4L2: Video4Linux2 is a collection of device drivers and API for supporting real-time video capture on Linux systems.

Support

Contact Us

If you need any support on e-CAM_YUV-GMSL-PRODUCTS product, please contact us using the Live Chat option available on our website - <https://www.e-consystems.com/>

Creating a Ticket

If you need to create a ticket for any type of issue, please visit the ticketing page on our website - <https://www.e-consystems.com/create-ticket.asp>

RMA

To know about our Return Material Authorization (RMA) policy, please visit the RMA Policy page on our website - <https://www.e-consystems.com/RMA-Policy.asp>

General Product Warranty Terms

To know about our General Product Warranty Terms, please visit the General Warranty Terms page on our website - <https://www.e-consystems.com/warranty.asp>

Revision History

Rev	Date	Description	Author
1.0	12-Aug-2024	Initial Draft	Camera Dev Team